

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

Problem Statement

An RL agent is trained for multiple episodes, and its reward increases gradually from **10 to 45**. Interpret this trend and explain what it indicates about the agent's learning performance and policy improvement.

Problem Solving Steps

1. Observe the reward values across episodes.
2. Compare the initial and final rewards.
3. Identify whether the reward is increasing or decreasing.
4. Calculate the improvement in reward.
5. Relate increasing reward to agent learning.
6. Check whether the policy is improving.

7. Identify the overall learning trend.
8. Conclude the agent's performance.

Program

```
import matplotlib.pyplot as plt
```

```
episodes = [1, 2, 3, 4, 5]
```

```
rewards = [10, 18, 27, 36, 45]
```

```
print("Initial Reward:", rewards[0])
```

```
print("Final Reward:", rewards[-1])
```

```
print("Reward Improvement:", rewards[-1] - rewards[0])
```

```
plt.plot(episodes, rewards, marker='o')
```

```
plt.xlabel("Episodes")
```

```
plt.ylabel("Reward")
```

```
plt.title("Reward Improvement over Episodes")
```

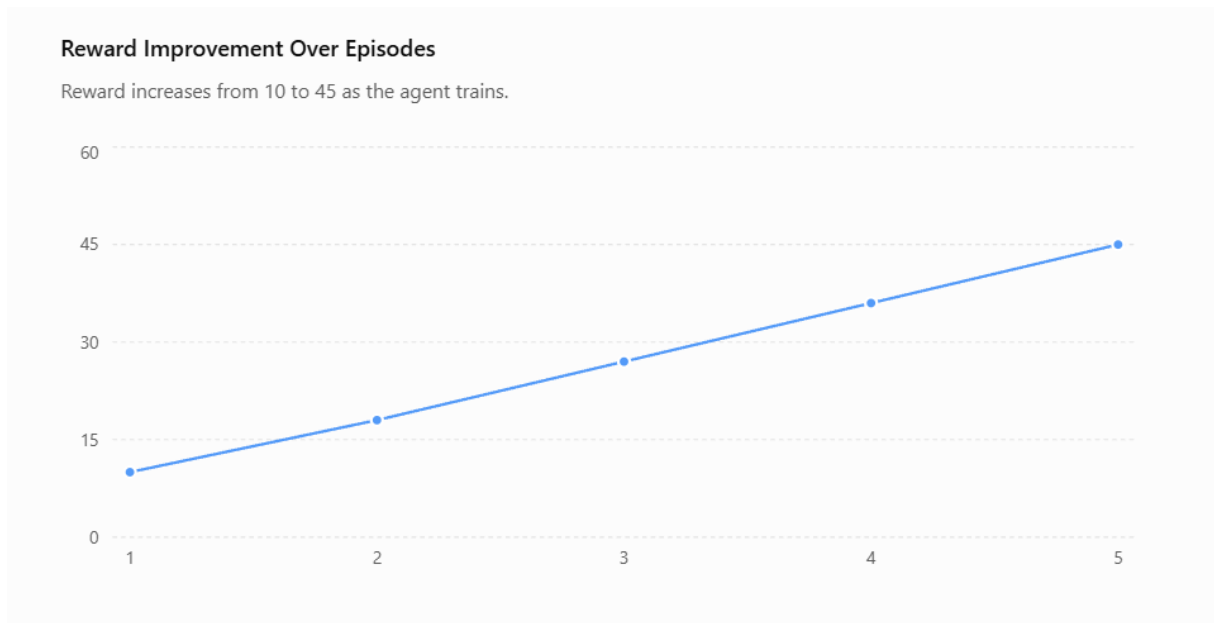
```
plt.grid()
```

```
plt.show()
```

```
plt.xlabel("Episodes")  
plt.ylabel("Reward")  
plt.title("Reward Improvement over Episodes")  
plt.grid()  
plt.show()
```

*** Initial Reward: 10
Final Reward: 45
Reward Improvement: 35





Conclusion

The reward increased from 10 to 45, showing that the agent is learning effectively. The increasing reward indicates policy improvement and better decision-making as training progresses.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

Problem Statement

An ϵ -greedy agent is tested with $\epsilon = 0.1, 0.3$, and 0.5 . The cumulative reward is highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$. Interpret the effect of exploration and exploitation and identify the best ϵ value.

Problem Solving Steps

1. Consider the three ϵ values.
2. Observe the cumulative rewards.
3. Understand that ϵ controls exploration.
4. Higher ϵ means more random exploration.
5. Lower ϵ means more exploitation of learned actions.
6. Compare the rewards for each ϵ .
7. Identify the ϵ value with the highest reward.
8. Conclude the best-performing value.

Program

```
import matplotlib.pyplot as plt
```

```
epsilon = [0.1, 0.3, 0.5]
```

```
reward = [90, 70, 50]
```

```
print("Best Epsilon:", epsilon[reward.index(max(reward))])
```

```
print("Highest Reward:", max(reward))
```

```
plt.plot(epsilon, reward, marker='o')
```

```
plt.xlabel("Epsilon")
```

```
plt.ylabel("Cumulative Reward")
```

```
plt.title("Epsilon vs Cumulative Reward")
```

```
plt.grid()
```

```
plt.show()
```

[2]
✓ 0s

```
import matplotlib.pyplot as plt

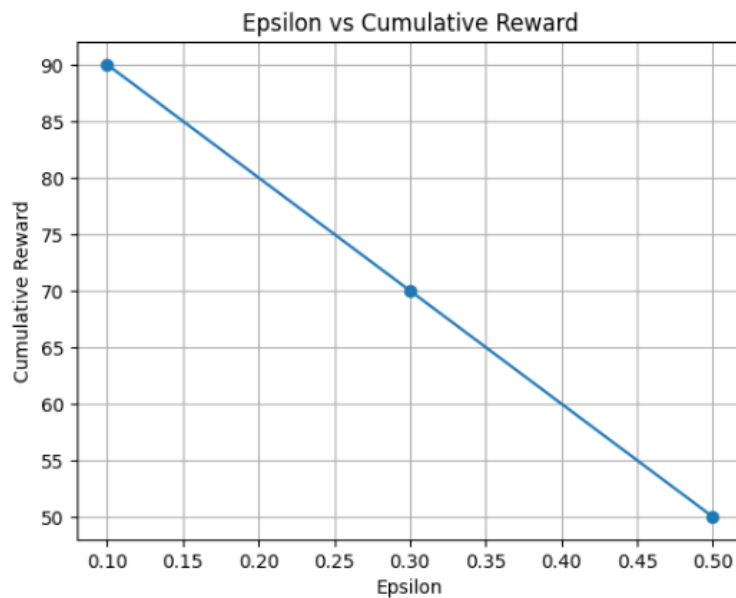
epsilon = [0.1, 0.3, 0.5]
reward = [90, 70, 50]

print("Best Epsilon:", epsilon[reward.index(max(reward))])
print("Highest Reward:", max(reward))

plt.plot(epsilon, reward, marker='o')
plt.xlabel("Epsilon")
plt.ylabel("Cumulative Reward")
plt.title("Epsilon vs Cumulative Reward")
plt.grid()
plt.show()
```

✓

... Best Epsilon: 0.1
Highest Reward: 90





Conclusion

$\epsilon = 0.1$ provides the best performance because the agent performs more exploitation and less random exploration, resulting in the highest cumulative reward

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)

Problem Statement

Compare Q-Learning and SARSA based on their average rewards over 100, 200, and 300 episodes, and determine which algorithm performs better and converges faster.

Problem Solving Steps

1. Define the episode values.
2. Record average rewards for Q-Learning.
3. Record average rewards for SARSA.
4. Compare rewards at each episode.
5. Observe the reward improvement.
6. Identify which algorithm has higher rewards.
7. Analyze the convergence trend.
8. Select the better-performing algorithm.

Program

```
import matplotlib.pyplot as plt
```

```
episodes = [100, 200, 300]
```

```
q_learning = [60, 75, 88]
```

```
sarsa = [55, 70, 82]
```

```
print("Q-Learning Rewards:", q_learning)
```

```
print("SARSA Rewards:", sarsa)
```

```
if q_learning[-1] > sarsa[-1]:
```

```
    print("Better Algorithm: Q-Learning")
```

```
else:
```

```
    print("Better Algorithm: SARSA")
```

```
plt.plot(epochs, q_learning, marker='o', label="Q-Learning")
```

```
plt.plot(epochs, sarsa, marker='o', label="SARSA")
```

```
plt.xlabel("Episodes")
```

```
plt.ylabel("Average Reward")
```

```
plt.title("Q-Learning vs SARSA")
```

```
plt.legend()
```

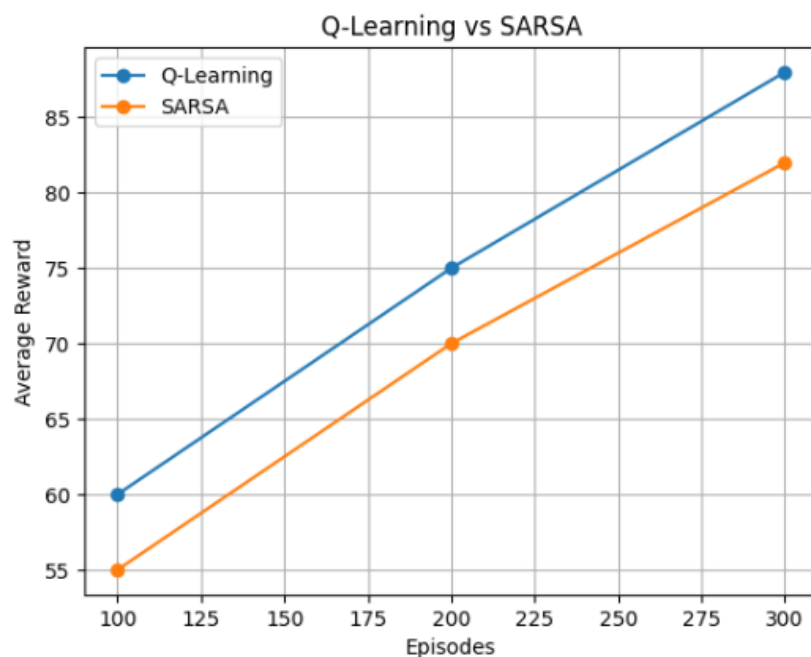
```
plt.grid()
```

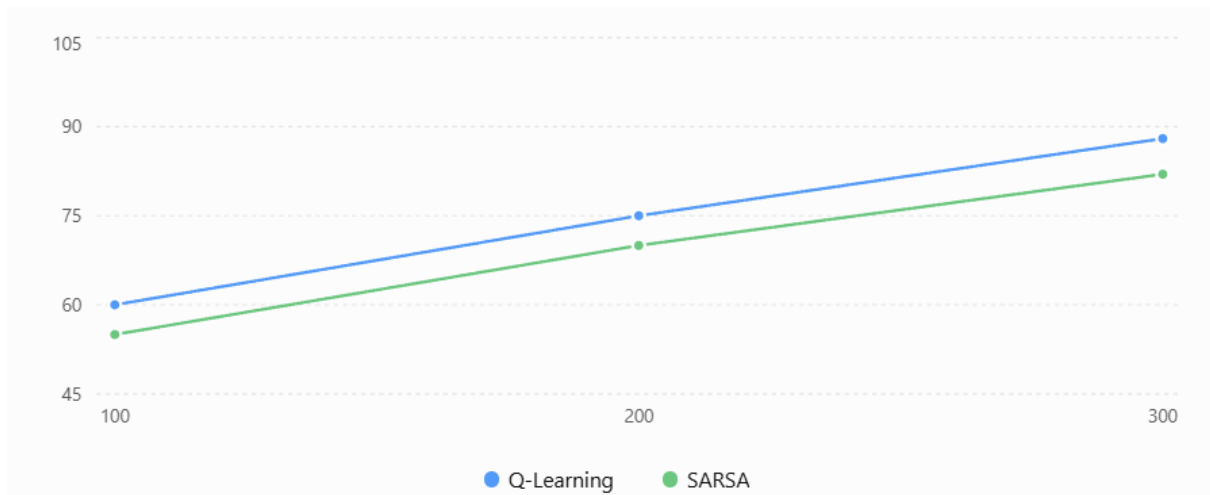
```
plt.show()
```

```
plt.ylabel("Average Reward")  
plt.title("Q-Learning vs SARSA")  
plt.legend()  
plt.grid()  
plt.show()
```

▼

```
*** Q-Learning Rewards: [60, 75, 88]  
SARSA Rewards: [55, 70, 82]  
Better Algorithm: Q-Learning
```





Conclusion

Q-Learning performs better than SARSA because it consistently gives slightly higher rewards. Based on the given trend, Q-Learning shows faster/better convergence toward higher reward values.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

Problem Statement

An RL agent is trained using different discount factor values 0.2, 0.5, and 0.9. The average reward increases as the discount factor increases. Analyze how γ affects long-term reward optimization and identify the most suitable value.

Problem Solving Steps

1. Define the discount factor values.
2. Record the average reward for each value.
3. Compare the obtained rewards.
4. Observe the reward trend.
5. Understand the effect of γ on future rewards.
6. Identify the highest reward.
7. Select the best discount factor.
8. Conclude its effect on learning.

Program

```
import matplotlib.pyplot as plt
```

```
gamma = [0.2, 0.5, 0.9]
reward = [40, 60, 85]
```

```
print("Discount Factors:", gamma)
print("Average Rewards:", reward)
```

```
best = gamma[reward.index(max(reward))]
```

```
print("Best Discount Factor:", best)
print("Highest Reward:", max(reward))
```

```
plt.plot(gamma, reward, marker='o')
plt.xlabel("Discount Factor ( $\gamma$ )")
plt.ylabel("Average Reward")
plt.title("Discount Factor vs Average Reward")
plt.grid()
plt.show()
```

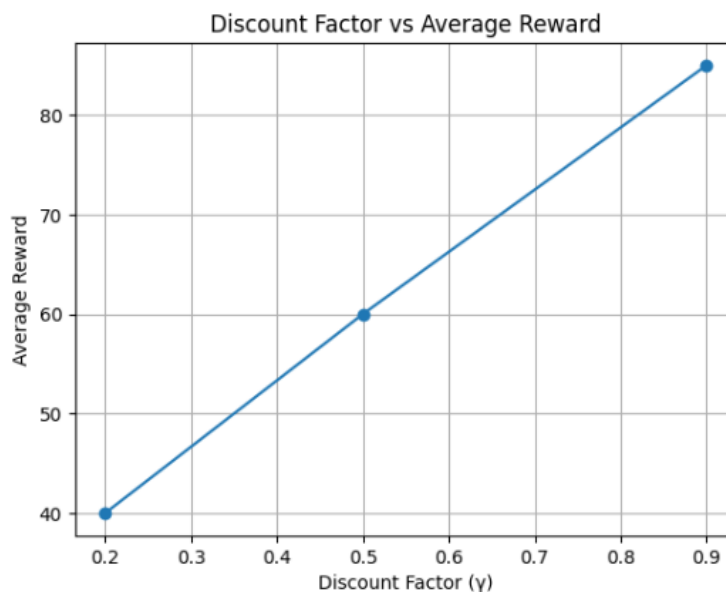
```
[5] print("Discount Factors:", gamma)
    print("Average Rewards:", reward)

    best = gamma[reward.index(max(reward))]

    print("Best Discount Factor:", best)
    print("Highest Reward:", max(reward))

    plt.plot(gamma, reward, marker='o')
    plt.xlabel("Discount Factor ( $\gamma$ )")
    plt.ylabel("Average Reward")
    plt.title("Discount Factor vs Average Reward")
    plt.grid()
    plt.show()
```

```
*** Discount Factors: [0.2, 0.5, 0.9]
    Average Rewards: [40, 60, 85]
    Best Discount Factor: 0.9
    Highest Reward: 85
```





Conclusion

The average reward increases with the discount factor. $\gamma = 0.9$ gives the highest reward, indicating better consideration of long-term rewards and better learning performance.

5. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

Problem Statement

An RL agent is trained in the CartPole environment using learning rates 0.01, 0.05, and 0.1. The agent achieves better stability with 0.05. Analyze how the learning rate affects convergence and identify the best trade-off between speed and stability.

Problem Solving Steps

1. Define the learning rate values.
2. Record the stability for each value.
3. Compare the learning performance.
4. Observe the convergence trend.
5. Check whether learning is stable.
6. Identify the most stable value.
7. Compare speed and stability.
8. Select the best learning rate.

Program

```
import matplotlib.pyplot as plt
```

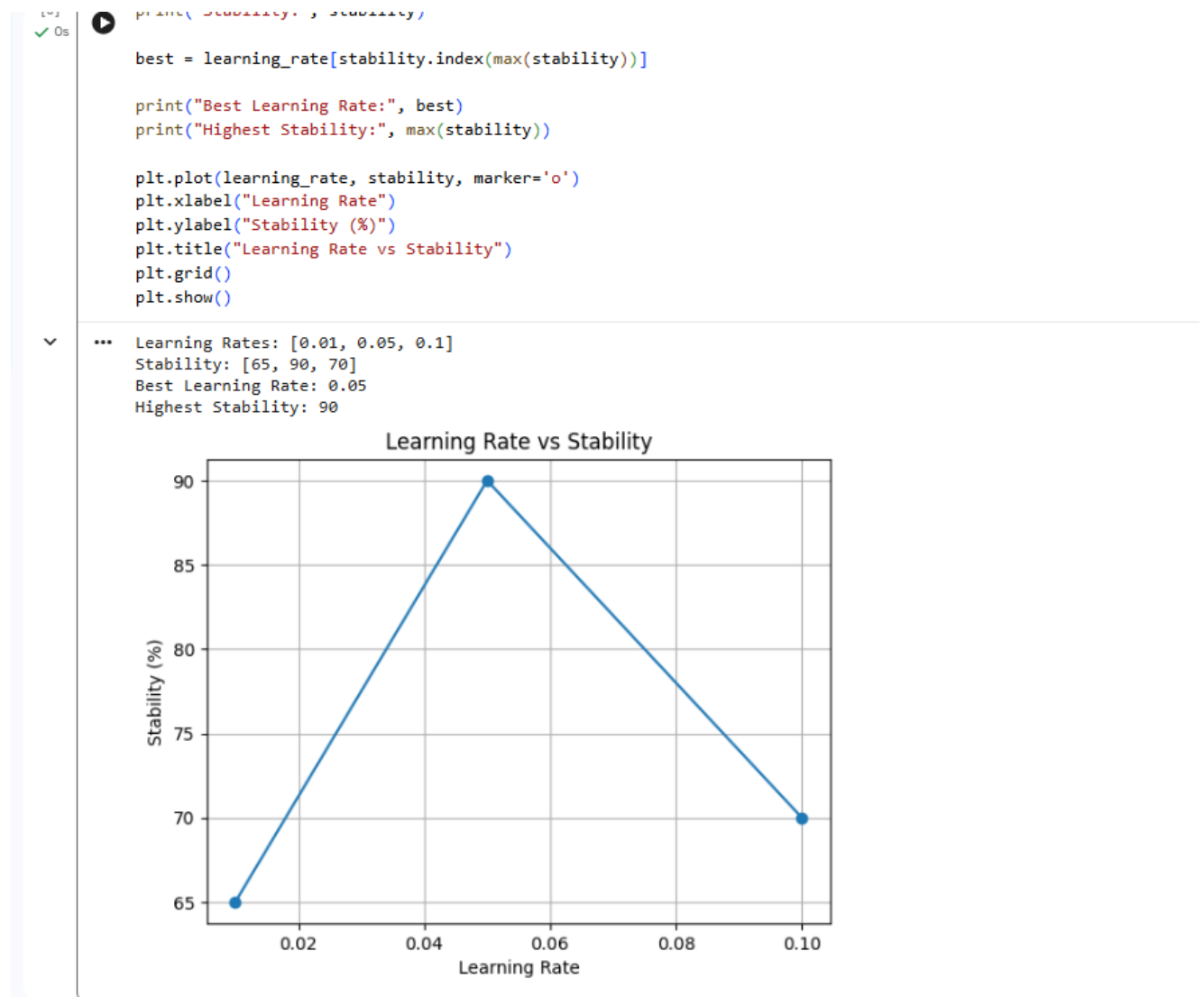
```
learning_rate = [0.01, 0.05, 0.1]  
stability = [65, 90, 70]
```

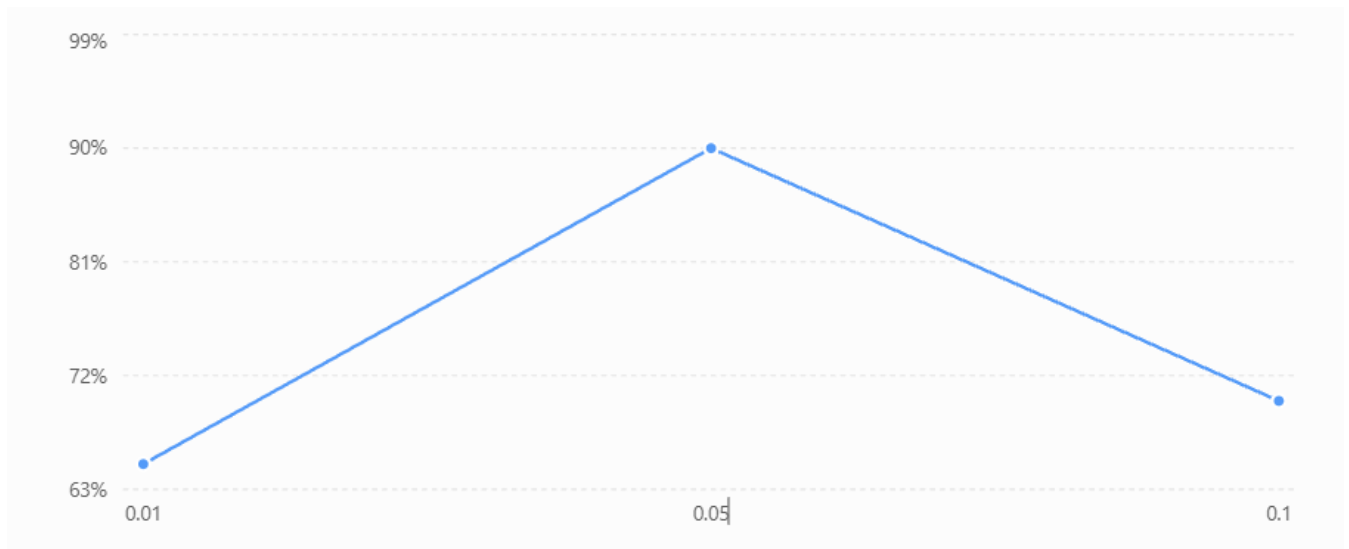
```
print("Learning Rates:", learning_rate)
print("Stability:", stability)
```

```
best = learning_rate[stability.index(max(stability))]
```

```
print("Best Learning Rate:", best)
print("Highest Stability:", max(stability))
```

```
plt.plot(learning_rate, stability, marker='o')
plt.xlabel("Learning Rate")
plt.ylabel("Stability (%)")
plt.title("Learning Rate vs Stability")
plt.grid()
plt.show()
```





Conclusion

A learning rate of 0.05 provides the best balance between convergence speed and stability. Very low or high learning rates result in lower stability.

Code of Conduct:

I M.Samanvi 192425318 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Samanvi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand